

Qatar-1b

R-1880

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_141108 · created 2026-08-03T11:42:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456969.67507 +/- 0.00041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.014
Transit depth (Rp/Rs)^2	2.47 +/- 0.45 [%]
Orbital Inclination (inc)	83.93 +/- 1.73
Airmass coefficient 1 (a1)	1.012 +/- 0.012
Airmass coefficient 2 (a2)	-0.0064 +/- 0.0097
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	#2 - [381, 117]
Optimal Method	PSF photometry
Transit Duration (day)	0.067 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.157 ± 0.014 vs 0.1463 (deviation 0.76σ)
Duration fit vs published	0.067 d vs 0.0692 d (deviation 0.18σ)
Mid-time O-C	-1.0 min
Depth SNR	11.2
Residual scatter	1.15 %

Light curve

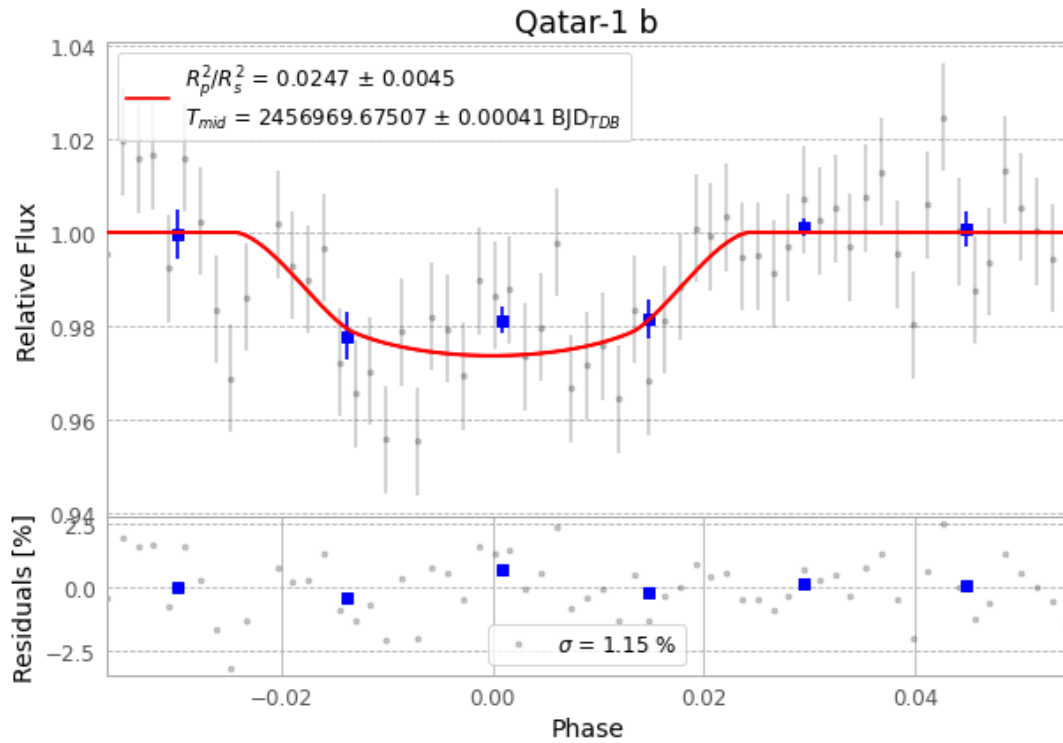


Figure 1 — FinalLightCurve_Qatar-1 b_2014-11-07.png (SHA-256 968b6a5a695eecab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 333950c565f466f1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), Qatar-1141108025412.FITS → Qatar-1141108060013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-141108.log (07be6bc35b10...), FinalLightCurve_Qatar-1 b_2014-11-07.png (968b6a5a695e...), FinalLightCurve_Qatar-1 b_2014-11-07.csv (03e49436698e...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:42Z.